Spring Batch — ExecutionContext

1. What is ExecutionContext?

ExecutionContext is a key-value storage area where Spring Batch can save the state or progress of a job or step. It is especially useful for restartability.

Package in Spring Batch 6

```
import org.springframework.batch.infrastructure.item.ExecutionContext;
```

2. Basic Example

```
ExecutionContext context = new ExecutionContext();

context.put("lastProcessedId", 100);
context.put("fileName", "customersrecords.csv");

int id = context.getInt("lastProcessedId");
String file = context.getString("fileName");
```

The context contains key-value pairs such as **lastProcessedId** → **100** and **fileName** → **customersrecords.csv**.

3. Why does Spring Batch need it?

The most important purpose is **restartability**. If a batch job fails after processing part of the data, Spring Batch can save execution state and use that state when the job is restarted.

```
Batch starts
    ↓
Process records
    ↓
Save progress
    ↓
ExecutionContext
    ↓
Application crashes
    ↓
Restart job
    ↓
Read saved state
    ↓
Continue processing
```

4. ExecutionContext and a CSV reader

A custom reader can save its position using ExecutionContext:

```
public void update(ExecutionContext executionContext) {
    executionContext.putLong("currentPosition", currentPosition);
}
```

When restarting, it can retrieve the saved position:

```
public void open(ExecutionContext executionContext) {
    long position = executionContext.getLong("currentPosition");
}
```

5. JobParameters vs ExecutionContext

JobParameters	ExecutionContext
Input provided when starting the job	State/progress saved during execution
Example: fileName, runId	Example: currentLine, lastProcessedCustomer
Used to identify/configure a job run	Used to maintain execution state and support restartability

6. Two levels of ExecutionContext

```
Job
 ■
 ■■■ JobExecution
       ■
       ■■■ Job ExecutionContext

       ■■■ StepExecution
             ■
             ■■■ Step ExecutionContext
```

Spring Batch has an execution context associated with a JobExecution and another associated with a StepExecution, allowing state to be maintained at different scopes.

7. Easy way to remember

```
JobParameters   → input provided when starting the job

ExecutionContext → state/progress saved during the job
```

One-line definition: ExecutionContext is a persistent key-value store used by Spring Batch to save execution state so that a job or step can be restarted from an appropriate point after a failure.